

JEE MAIN REVISION MASTER: DIFFERENTIATION

2026 4 QUESTIONS 6.67% WEIGHTAGE $\uparrow$ 33.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Differentiation - Core Application	38 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Differentiation - Core Application (Very High Yield — 38 PYQs)

Core Principle: Key problem pattern in Kcet Differentiation.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The second order derivative of $\cos^{-1}(4x^3-3x)$ with respect to $\cos^{-1}(2x^2-1)$, where $(1/2) < x < 1$ is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If $f(x) = \sin^{-1}(\frac{2x+1}{x^2+1})$, then $f'(\frac{1}{2}) = \dots$ " [Ans: A]
- [2026 Shift] Q#3 (MCQ): "If $y = \sqrt{3}$, then $(dy/dx) = \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The second order derivative of $\cos^{-1}(4x^3-3x)$ with respect to $\cos^{-1}(2x^2-1)$, where $(1/2) < x < 1$ is

- (A) 0
- (B) $-\frac{1}{\sqrt{1-x^2}}$
- (C) $(3/2)$
- (D) $-(3/2)$

Correct Answer: None

Solution Analysis: Take $y = \cos^{-1}(4x^3-3x)$ and $t = \cos^{-1}(2x^2-1)$ We need (d^2y / dt^2) Since $(1/2) < x < 1$ where $0 < \theta < (\pi/3)$ Now use the standard identities: $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ and $2\cos^2\theta - 1 = \cos 2\theta$ So, $y = \cos^{-1}(\cos 3\theta)$ Since $0 < 3\theta < \pi$, we get $y = 3\theta$ Also, $t = \cos^{-1}(\cos 2\theta)$ Since $0 < 2\theta < \pi$, we get $t = 2\theta$ Thus, $y = 3\theta$, $t = 2\theta$ So, $y = (3/2)t$ Hence, $(dy / dt) = (3/2)$ and $(d^2y / dt^2) = 0$ So the correct o